16.2 Singularities and Zeros.

Roughly, a *singular point* of an analytic function $f(z)$ is a z_0 at which $f(z)$ ceases to be analytic, and a *zero* is a z at which $f(z) = 0$. Precise definitions follow below. In this section we show that Laurent series can be used for classifying singularities and Taylor series for discussing zeros.

Singularities were defined in Sec. 15.4, as we shall now recall and extend. We also remember that, by definition, a function is a ***single-valued*** relation, as was emphasized in Sec. 13.3.

We say that a function $f(z)$ **is singular** or **has a singularity** at a point $z = z_0$ if $f(z)$ is not analytic (perhaps not even defined) at $z = z_0$, but every neighborhood of $z = z_0$ contains points at which $f(z)$ is analytic. We also say that $z = z_0$ is a **singular point** of $f(z)$.

We call $z = z_0$ an **isolated singularity** of $f(z)$ if $z = z_0$ has a neighborhood without further singularities of $f(z)$. *Example:* $\tan z$ has isolated singularities at $\pm\pi/2$, $\pm 3\pi/2$, etc.; $\tan(1/z)$ has a nonisolated singularity at 0. (Explain!)

Isolated singularities of $f(z)$ at $z = z_0$ can be classified by the Laurent series

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n} \tag{1}$$

(Sec. 16.1)

valid ***in the immediate neighborhood*** of the singular point $z = z_0$, except at z_0 itself, that is, in a region of the form

$$0 < |z - z_0| < R.$$

The sum of the first series is analytic at $z = z_0$, as we know from the last section. The second series, containing the negative powers, is called the **principal part** of (1), as we remember from the last section. If it has only finitely many terms, it is of the form

$$\frac{b_1}{z - z_0} + \cdots + \frac{b_m}{(z - z_0)^m} \qquad (b_m \neq 0). \tag{2}$$

Then the singularity of $f(z)$ at $z = z_0$ is called a **pole**, and m is called its **order**. Poles of the first order are also known as **simple poles**.

If the principal part of (1) has infinitely many terms, we say that $f(z)$ has at $z = z_0$ an **isolated essential singularity**.

We leave aside nonisolated singularities.

EXAMPLE 1

Poles. Essential Singularities

The function

$$f(z) = \frac{1}{z(z-2)^5} + \frac{3}{(z-2)^2}$$

has a simple pole at $z = 0$ and a pole of fifth order at $z = 2$. Examples of functions having an isolated essential singularity at $z = 0$ are

$$e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n!z^n} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \cdots$$

and

$$\sin\frac{1}{z} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!z^{2n+1}} = \frac{1}{z} - \frac{1}{3!z^3} + \frac{1}{5!z^5} - + \cdots.$$

Section 16.1 provides further examples. In that section, Example 1 shows that $z^{-5}\sin z$ has a fourth-order pole at 0. Furthermore, Example 4 shows that $1/(z^3 - z^4)$ has a third-order pole at 0 and a Laurent series with infinitely many negative powers. This is no contradiction, since this series is valid for $|z| > 1$; it merely tells us that in classifying singularities it is quite important to consider the Laurent series valid ***in the immediate neighborhood*** of a singular point. In Example 4 this is the series (I), which has three negative powers.

Removable Singularities. We say that a function $f(z)$ has a *removable singularity* at $z = z_0$ if $f(z)$ is not analytic at $z = z_0$, but can be made analytic there by assigning a suitable value $f(z_0)$. Such singularities are of no interest since they can be removed as just indicated. *Example:* $f(z) = (\sin z)/z$ becomes analytic at $z = 0$ if we define $f(0) = 1$.

Zeros of Analytic Functions

A **zero** of an analytic function $f(z)$ in a domain D is a $z = z_0$ in D such that $f(z_0) = 0$. A zero has **order** n if not only f but also the derivatives $f', f'', \cdots, f^{(n-1)}$ are all 0 at $z = z_0$ but $f^{(n)}(z_0) \neq 0$. A first-order zero is also called a **simple zero**. For a second-order zero, $f(z_0) = f'(z_0) = 0$ but $f''(z_0) \neq 0$. And so on.

EXAMPLE 4

Zeros

The function $1 + z^2$ has simple zeros at $\pm i$. The function $(1 - z^4)^2$ has second-order zeros at ± 1 and $\pm i$. The function $(z - a)^3$ has a third-order zero at $z = a$. The function e^z has no zeros (see Sec. 13.5). The function $\sin z$ has simple zeros at $0, \pm\pi, \pm 2\pi, \cdots$, and $\sin^2 z$ has second-order zeros at these points. The function $1 - \cos z$ has second-order zeros at $0, \pm 2\pi, \pm 4\pi, \cdots$, and the function $(1 - \cos z)^2$ has fourth-order zeros at these points. ■

Taylor Series at a Zero. At an nth-order zero $z = z_0$ of $f(z)$, the derivatives $f'(z_0), \cdots, f^{(n-1)}(z_0)$ are zero, by definition. Hence the first few coefficients $a_0, \cdots, a_{n-1}$ of the Taylor series (1), Sec. 15.4, are zero, too, whereas $a_n \neq 0$, so that this series takes the form

$$
\begin{aligned}
f(z) &= a_n(z - z_0)^n + a_{n+1}(z - z_0)^{n+1} + \cdots \\
&= (z - z_0)^n \left[a_n + a_{n+1}(z - z_0) + a_{n+2}(z - z_0)^2 + \cdots\right] \qquad (a_n \neq 0).
\end{aligned} \tag{3}
$$

This is characteristic of such a zero, because, if $f(z)$ has such a Taylor series, it has an nth-order zero at $z = z_0$, as follows by differentiation.

Whereas nonisolated singularities may occur, for zeros we have

THEOREM 3

Zeros

The zeros of an analytic function $f(z)$ $(\not\equiv 0)$ are isolated; that is, each of them has a neighborhood that contains no further zeros of $f(z)$.

16.3 Residue Integration Method

We now cover a second method of evaluating complex integrals. Recall that we solved complex integrals directly by Cauchy's integral formula in Sec. 14.3. In Chapter 15 we learned about power series and especially Taylor series. We generalized Taylor series to Laurent series (Sec. 16.1) and investigated singularities and zeroes of various functions (Sec. 16.2). Our hard work has paid off and we see how much of the theoretical groundwork comes together in evaluating complex integrals by the residue method.

The purpose of Cauchy's residue integration method is the evaluation of integrals

$$\oint_C f(z)\, dz$$

taken around a simple closed path C. The idea is as follows.

If $f(z)$ is analytic everywhere on C and inside C, such an integral is zero by Cauchy's integral theorem (Sec. 14.2), and we are done.

The situation changes if $f(z)$ has a singularity at a point $z = z_0$ inside C but is otherwise analytic on C and inside C as before. Then $f(z)$ has a Laurent series

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \frac{b_1}{z - z_0} + \frac{b_2}{(z - z_0)^2} + \cdots$$

that converges for all points near $z = z_0$ (except at $z = z_0$ itself), in some domain of the form $0 < |z - z_0| < R$ (sometimes called a **deleted neighborhood**, an old-fashioned term that we shall not use). Now comes the key idea. The coefficient b_1 of the first negative power $1/(z - z_0)$ of this Laurent series is given by the integral formula (2) in Sec. 16.1 with $n = 1$, namely,

$$b_1 = \frac{1}{2\pi i} \oint_C f(z)\, dz.$$

Now, since we can obtain Laurent series by various methods, without using the integral formulas for the coefficients (see the examples in Sec. 16.1), we can find b_1 by one of those methods and then use the formula for b_1 for evaluating the integral, that is,

$$\oint_C f(z)\, dz = 2\pi i b_1. \tag{1}$$

Here we integrate counterclockwise around a simple closed path C that contains $z = z_0$ in its interior (but no other singular points of $f(z)$ on or inside C!).

The coefficient b_1 is called the **residue** of $f(z)$ at $z = z_0$ and we denote it by

$$b_1 = \operatorname*{Res}_{z=z_0} f(z). \tag{2}$$

EXAMPLE 1

Evaluation of an Integral by Means of a Residue

Integrate the function $f(z) = z^{-4} \sin z$ counterclockwise around the unit circle C.

Solution. From (14) in Sec. 15.4 we obtain the Laurent series

$$f(z) = \frac{\sin z}{z^4} = \frac{1}{z^3} - \frac{1}{3!z} + \frac{1}{5!} - \frac{z^3}{7!} + - \cdots$$

which converges for $|z| > 0$ (that is, for all $z \neq 0$). This series shows that $f(z)$ has a pole of third order at $z = 0$ and the residue $b_1 = -\frac{1}{3}!$. From (1) we thus obtain the answer

$$\oint_C \frac{\sin z}{z^4}\, dz = 2\pi i b_1 = -\frac{\pi i}{3}.$$ ■

Example # 01 $\quad$ 2nd Method

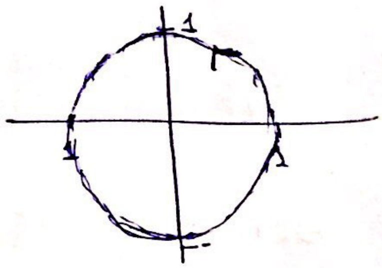
Integrate the function $f(z) = z^{-4}\sin z$ counterclockwise around the unit circle c.

Sol:-

$$\oint_c z^{-4}\sin z \, dz \qquad \text{where} \qquad c: |z| = 1$$

$$\oint_c \frac{\sin z}{z^4} dz$$

$\downarrow$ not analytic at $z = 0$



By Derivative of the analytic ftn we have.

$$\oint_c \frac{f(z)}{(z-z_0)^{n+1}} dz = \frac{2\pi i}{n!} f^n(z_0) \longrightarrow (*)$$

Here $n = 3$, $z_0 = 0$, $f(z) = \sin z$

$$\oint_c \frac{\sin z}{z^4} dz = \frac{2\pi i}{3!} f^{3}(0) \longrightarrow (1)$$

$f(z) = \sin z$	$f'''(z) = -\cos z$
$f'(z) = \cos z$	$f'''(0) = -\cos(0) = -1$
$f''(z) = -\sin z$	from (1)

$$\oint_c \sin z/z^4 \, dz = -\frac{\cancel{2}\pi i}{\cancel{6}3} = -\frac{\pi i}{3} \text{ Ans}$$

EXAMPLE 2

CAUTION! Use the Right Laurent Series!

Integrate $f(z) = 1/(z^3 - z^4)$ clockwise around the circle C: $|z| = \frac{1}{2}$.

Solution. $z^3 - z^4 = z^3(1 - z)$ shows that $f(z)$ is singular at $z = 0$ and $z = 1$. Now $z = 1$ lies outside C. Hence it is of no interest here. So we need the residue of $f(z)$ at 0. We find it from the Laurent series that converges for $0 < |z| < 1$. This is series (I) in Example 4, Sec. 16.1,

$$\frac{1}{z^3 - z^4} = \frac{1}{z^3} + \frac{1}{z^2} + \frac{1}{z} + 1 + z + \cdots \qquad (0 < |z| < 1).$$

We see from it that this residue is 1. Clockwise integration thus yields

$$\oint_C \frac{dz}{z^3 - z^4} = -2\pi i \operatorname*{Res}_{z=0} f(z) = -2\pi i.$$

CAUTION! Had we used the wrong series (II) in Example 4, Sec. 16.1,

$$\frac{1}{z^3 - z^4} = -\frac{1}{z^4} - \frac{1}{z^5} - \frac{1}{z^6} - \cdots \qquad (|z| > 1),$$

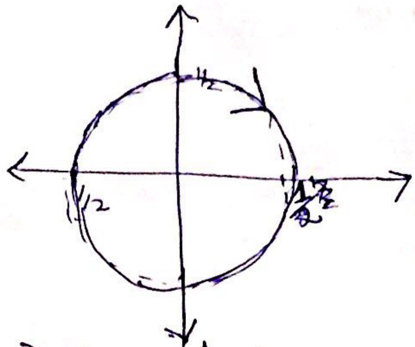
we would have obtained the wrong answer, 0, because this series has no power $1/z$. ■

Example # 2 $\quad$ 2nd Method

Integrate $f(z) = \dfrac{1}{z^3 - z^4}$ clockwise around the circle $C: |z| = \frac{1}{2}$

Sol:-

$$\oint_C \frac{1}{z^3(1-z)}\, dz$$



It is not analytic at $z=0$ and $z=1$

but $z=1$ lies outside the circle

$$\oint_C \frac{\frac{1}{1-z}}{z^3}\, dz$$

By using Derivative of the analytic ftn we have.

$$\oint_C \frac{f(z)}{(z-z_0)^{n+1}}\, dz = \frac{2\pi i}{n!} f^n(z_0) \longrightarrow (*)$$

Here $\boxed{n=2}$, $\boxed{z_0=0}$

$$\oint_C \frac{\frac{1}{1-z}}{z^3}\, dz = \frac{2\pi i}{2!} f^2(z_0) \longrightarrow (1)$$

$f(z) = (1-z)^{-1}$

$f'(z) = (-1)(1-z)^{-2}(-1) = (1-z)^{-2}$

$f''(z) = (-2)(1-z)^{-3}(-1)$

$f''(z) = 2(1-z)^{-3}$

$$\boxed{f''(0) = 2}$$

from 1

$$\oint_c \frac{\frac{1}{1-z}}{z^3} dz = \frac{2\pi i}{\cancel{2}}(\cancel{2}) = -2\pi i$$

(-) sign is b/s of clockwise

Formulas for Residues

To calculate a residue at a pole, we need not produce a whole Laurent series, but, more economically, we can derive formulas for residues once and for all.

Simple Poles at z_0. A first formula for the residue at a simple pole is

$$\operatorname*{Res}_{z=z_0} f(z) = b_1 = \lim_{z\to z_0} (z - z_0)f(z). \tag{3}$$

(Proof below).

A second formula for the residue at a simple pole is

$$\operatorname*{Res}_{z=z_0} f(z) = \operatorname*{Res}_{z=z_0} \frac{p(z)}{q(z)} = \frac{p(z_0)}{q'(z_0)}. \tag{4}$$

(Proof below).

In (4) we assume that $f(z) = p(z)/q(z)$ with $p(z_0) \neq 0$ and $q(z)$ has a simple zero at z_0, so that $f(z)$ has a simple pole at z_0 by Theorem 4 in Sec. 16.2.

PROOF We prove (3). For a simple pole at $z = z_0$ the Laurent series (1), Sec. 16.1, is

$$f(z) = \frac{b_1}{z - z_0} + a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots \qquad (0 < |z - z_0| < R).$$

Here $b_1 \neq 0$. (Why?) Multiplying both sides by $z - z_0$ and then letting $z \to z_0$, we obtain the formula (3):

$$\lim_{z\to z_0} (z - z_0)f(z) = b_1 + \lim_{z\to z_0} (z - z_0)[a_0 + a_1(z - z_0) + \cdots] = b_1$$

where the last equality follows from continuity (Theorem 1, Sec. 15.3).

We prove (4). The Taylor series of $q(z)$ at a simple zero z_0 is

$$q(z) = (z - z_0)q'(z_0) + \frac{(z - z_0)^2}{2!} q''(z_0) + \cdots.$$

Substituting this into $f = p/q$ and then f into (3) gives

$$\operatorname*{Res}_{z=z_0} f(z) = \lim_{z\to z_0} (z - z_0)\frac{p(z)}{q(z)} = \lim_{z\to z_0} \frac{(z - z_0)p(z)}{(z - z_0)[q'(z_0) + (z - z_0)q''(z_0)/2 + \cdots]}.$$

$z - z_0$ cancels. By continuity, the limit of the denominator is $q'(z_0)$ and (4) follows. ■

EXAMPLE 3

Residue at a Simple Pole

$f(z) = (9z + i)/(z^3 + z)$ has a simple pole at i because $z^2 + 1 = (z + i)(z - i)$, and (3) gives the residue

$$\operatorname*{Res}_{z=i} \frac{9z + i}{z(z^2 + 1)} = \lim_{z \to i} (z - i) \frac{9z + i}{z(z + i)(z - i)} = \left[\frac{9z + i}{z(z + i)}\right]_{z=i} = \frac{10i}{-2} = -5i.$$

By (4) with $p(i) = 9i + i$ and $q'(z) = 3z^2 + 1$ we confirm the result,

$$\operatorname*{Res}_{z=i} \frac{9z + i}{z(z^2 + 1)} = \left[\frac{9z + i}{3z^2 + 1}\right]_{z=i} = \frac{10i}{-2} = -5i.$$

Poles of Any Order at z_0. The residue of $f(z)$ at an mth-order pole at z_0 is

$$\operatorname*{Res}_{z=z_0} f(z) = \frac{1}{(m - 1)!} \lim_{z \to z_0} \left\{\frac{d^{m-1}}{dz^{m-1}} \left[(z - z_0)^m f(z)\right]\right\}. \tag{5}$$

In particular, for a second-order pole ($m = 2$),

$$\operatorname*{Res}_{z=z_0} f(z) = \lim_{z \to z_0} \{[(z - z_0)^2 f(z)]'\}. \tag{5*}$$

EXAMPLE 4

Residue at a Pole of Higher Order

$f(z) = 50z/(z^3 + 2z^2 - 7z + 4)$ has a pole of second order at $z = 1$ because the denominator equals $(z + 4)(z - 1)^2$ (verify!). From (5*) we obtain the residue

$$\operatorname*{Res}_{z=1} f(z) = \lim_{z \to 1} \frac{d}{dz} [(z - 1)^2 f(z)] = \lim_{z \to 1} \frac{d}{dz} \left(\frac{50z}{z + 4}\right) = \frac{200}{5^2} = 8.$$

Several Singularities Inside the Contour. Residue Theorem

Residue integration can be extended from the case of a single singularity to the case of several singularities within the contour C. This is the purpose of the residue theorem. The extension is surprisingly simple.

THEOREM 1

Residue Theorem

Let $f(z)$ be analytic inside a simple closed path C and on C, except for finitely many singular points $z_1, z_2, \cdots, z_k$ inside C. Then the integral of $f(z)$ taken counterclockwise around C equals $2\pi i$ times the sum of the residues of $f(z)$ at $z_1, \cdots, z_k$:

$$\oint_C f(z)\, dz = 2\pi i \sum_{j=1}^{k} \operatorname*{Res}_{z=z_j} f(z). \tag{6}$$

EXAMPLE 5

Integration by the Residue Theorem. Several Contours

Evaluate the following integral counterclockwise around any simple closed path such that (a) 0 and 1 are inside C, (b) 0 is inside, 1 outside, (c) 1 is inside, 0 outside, (d) 0 and 1 are outside.

$$\oint_C \frac{4-3z}{z^2-z}\, dz$$

Solution. The integrand has simple poles at 0 and 1, with residues [by (3)]

$$\operatorname*{Res}_{z=0} \frac{4-3z}{z(z-1)} = \left[\frac{4-3z}{z-1}\right]_{z=0} = -4, \qquad \operatorname*{Res}_{z=1} \frac{4-3z}{z(z-1)} = \left[\frac{4-3z}{z}\right]_{z=1} = 1.$$

[Confirm this by (4).] *Answer:* (a) $2\pi i(-4+1) = -6\pi i$, (b) $-8\pi i$, (c) $2\pi i$, (d) 0.

EXAMPLE 6

Another Application of the Residue Theorem

Integrate $(\tan z)/(z^2 - 1)$ counterclockwise around the circle C: $|z| = \frac{3}{2}$.

Solution. $\tan z$ is not analytic at $\pm\pi/2, \pm 3\pi/2, \cdots$, but all these points lie outside the contour C. Because of the denominator $z^2 - 1 = (z-1)(z+1)$ the given function has simple poles at ± 1. We thus obtain from (4) and the residue theorem

$$\oint_C \frac{\tan z}{z^2 - 1}\,dz = 2\pi i\left(\operatorname*{Res}_{z=1} \frac{\tan z}{z^2-1} + \operatorname*{Res}_{z=-1} \frac{\tan z}{z^2-1}\right)$$

$$= 2\pi i\left(\frac{\tan z}{2z}\bigg|_{z=1} + \frac{\tan z}{2z}\bigg|_{z=-1}\right)$$

$$= 2\pi i \tan 1 = 9.7855i.$$

Example 7

Evaluate the following integral, where C is the ellipse $9x^2 + y^2 = 9$ (counterclockwise, Sketch it).

$$\oint_C \left(\underbrace{\frac{z e^{\pi z}}{z^4 - 16}}_{I_1} + \underbrace{z e^{\pi / z}}_{I_2} \right) dz$$

Sol:-

Since $z^4 - 16 = 0$ at $\pm 2i, \pm 2$,

So the 1st term of the integrand has simple poles at $\pm 2i$ inside C,

$\because \; 9x^2 + y^2 = 9$

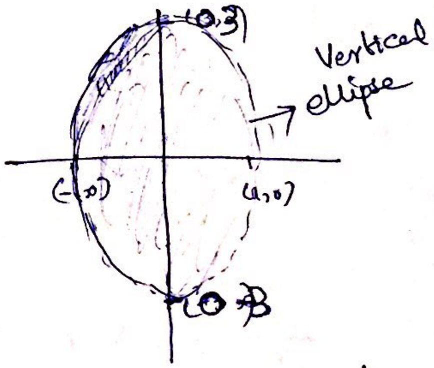
$\frac{9x^2}{9} + \frac{y^2}{9} = 1$

$\frac{x^2}{(1)^2} + \frac{y^2}{(3)^2} = 1$

$\because$ vertical Major axis

$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$

if $a > b$



So

$$\operatorname*{Res}_{z=2i} \frac{P(z)}{Q(z)} = \frac{P(z_0)}{Q'(z_0)}$$

So

$$\operatorname*{Res}_{z=2i} \left[\frac{z e^{\pi z}}{z^4 - 16} \right] = \left. \frac{\cancel{z} e^{\pi z}}{4 z^{\cancel{3}2}} \right|_{z=2i}$$

$$\Rightarrow \frac{e^{\pi z}}{4z^2} = \frac{e^{\pi (2i)}}{4(-4)} = \frac{e^{2\pi i}}{-16}$$

$$\Rightarrow -\frac{1}{16}$$

$\because$ $e^{2\pi i} = \cos 2\pi + i \sin 2\pi$

$\cos(2\pi) = (-1)^2$.

$$\operatorname*{Res}_{z=-2i} \frac{z e^{\pi z}}{z^4-16} = \left.\frac{z e^{\pi z}}{4z^3}\right|_{z=-2i}$$

$$\Rightarrow \left.\frac{e^{\pi z}}{4z^2}\right|_{z=-2i} = \frac{e^{\pi(-2i)}}{4(-2i)^2}$$

$$\Rightarrow \frac{e^{-i2\pi}}{4(+4i^2)} \Rightarrow \frac{e^{-2\pi i}}{-16} = \frac{-1}{16}$$

$e^{-2\pi i} = \cos 2\pi - i\sin 2\pi$

$\Rightarrow (-1)^2$

$= 1$

and simple poles at ± 2, which lie outside c, so that they are of no interest here. The second term has an essential singularity at 0, with residue $\frac{\pi^2}{2}$ as obtained from

$$z e^{\pi/z} = z\left[1 + \frac{\pi}{z} + \frac{\pi^2}{2!\,z^2} + \frac{\pi^3}{3!\,z^3} \cdots\right]$$

$$\Rightarrow z + \pi + \underbrace{\frac{\pi^2}{2!\,z}}_{b_1 \times \frac{1}{z}} + \frac{\pi^3}{3!\,z^2} + \cdots$$

So $\boxed{b_1 = \frac{\pi^2}{2!}}$

$$\oint_c \left(\underbrace{\frac{z e^{\pi z}}{z^4-16}}_{f(z)} + \underbrace{z e^{\pi/z}}_{g(z)}\right) dz = 2\pi i\left[\operatorname*{Res}_{z=2i} f(z) + \operatorname*{Res}_{z=2i} f(z) + \operatorname*{\cancel{Res}}_{z} b_1\right]$$

$$= 2\pi i\left[-\frac{1}{16} - \frac{1}{16} + \frac{\pi^2}{2}\right] = \pi\left(\pi^2 - \frac{1}{4}\right)i$$

$= 30.221i$ by Residue theorem

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